EASY ALGEBRA

Systems of two linear equations in two variables

03

10 questions · ~15 min

SAT QUESTION BANK 2026-05-29

Q1 ALGEBRA

Hiro and Sofia purchased shirts and pants from a store. The price of each shirt purchased was the same and the price of each pair of pants purchased was the same. Hiro purchased 4 shirts and 2 pairs of pants for \$86, and Sofia purchased 3 shirts and 5 pairs of pants for \$166. Which of the following systems of linear equations represents the situation, if x represents the price, in dollars, of each shirt and y represents the price, in dollars, of each pair of pants?

A. $4x+2y=86$
$3x+5y=166$

B. $4x+3y=86$
$2x+5y=166$

C. $4x+2y=166$
$3x+5y=86$

D. $4x+3y=166$
$2x+5y=86$

Q2

ALGEBRA

$$y = 4x - 9$$
$$y = 19$$

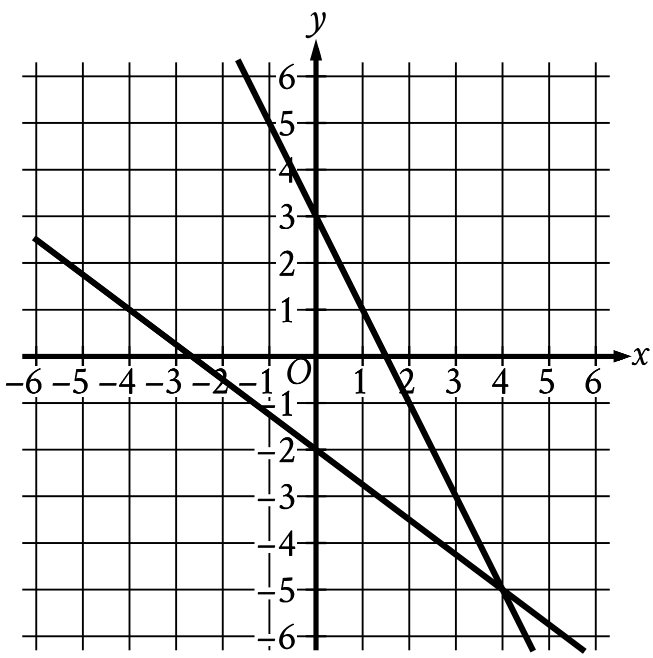
What is the solution x, y to the given system of equations?

A. $4, 19$

B. $7, 19$

C. $19, 4$

D. $19, 7$



- For the first line in the system:
  - The line slants sharply down from left to right.
  - The line passes through the following points:
    - (0 comma 3)
    - (4 comma negative 5)
- For the second line in the system:
  - The line slants gradually down from left to right.
  - The line passes through the following points:
    - (0 comma negative 2)
    - (4 comma negative 5)

The graph of a system of linear equations is shown. What is the solution x, y to the system?

A. $4, -5$

B. $0, 3$

C. $0, -2$

D. $-2, 3$

Q4 ALGEBRA

$$x + y = 20$$
$$2(x + y) + 3y = 85$$

If (x, y) is the solution to the given system of equations, what is the value of y ?

A. 10

B. 15

C. 60

D. 65

Q5 ALGEBRA

$$y = 12x - 20$$

$$y = 28$$

What is the solution x, y to the given system of equations?

A. $4, 28$

B. $20, 28$

C. $28, 4$

D. $28, 20$

Q6 ALGEBRA

$$10x = 110$$

$$6x - 63 = y$$

The solution to the given system of equations is x, y. What is the value of y?

A. 63

B. 11

C. 10

D. 3

Q7 ALGEBRA

A petting zoo sells two types of tickets. The standard ticket, for admission only, costs \$5. The premium ticket, which includes admission and food to give to the animals, costs \$12. One Saturday, the petting zoo sold a total of 250 tickets and collected a total of \$2,300 from ticket sales. Which of the following systems of equations can be used to find the number of standard tickets, s, and premium tickets, p, sold on that Saturday?

A. $$\begin{aligned} s + p &= 250 \\ 5s + 12p &= 2{,}300 \end{aligned}$$

B. $$\begin{aligned} s + p &= 250 \\ 12s + 5p &= 2{,}300 \end{aligned}$$

C. $$\begin{aligned} 5s + 12p &= 250 \\ s + p &= 2{,}300 \end{aligned}$$

D. $$\begin{aligned} 12s + 5p &= 250 \\ s + p &= 2{,}300 \end{aligned}$$

Q8

ALGEBRA

$$x = 4$$

$$y = 5 - x$$

The solution to the given system of equations is x, y. What is the value of y?

A. 1

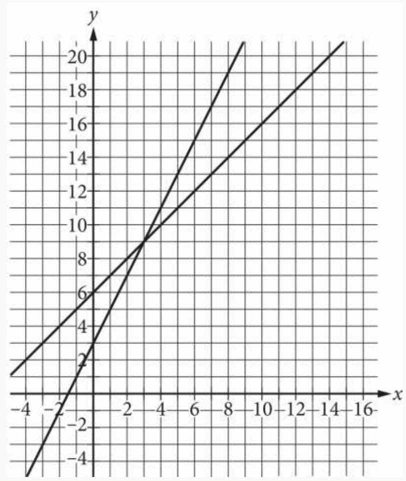
B. 4

C. 5

D. 9

Q9 ALGEBRA

A system of two linear equations is graphed in the xy-plane below.



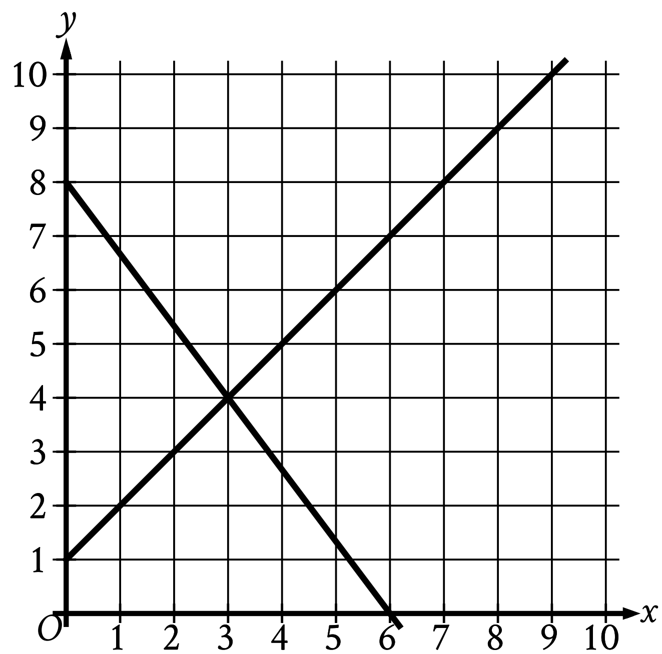
Which of the following points is the solution to the system of equations?

A. $(3,9)$

B. $(6,15)$

C. $(8,10)$

D. $(12,18)$



- For the first line in the system:
  - The line slants sharply down from left to right.
  - The line passes through the following points:
    - (0 comma 8)
    - (3 comma 4)
    - (5 comma four thirds)
- For the second line in the system:
  - The line slants gradually up from left to right.
  - The line passes through the following points:
    - (0 comma 1)
    - (3 comma 4)
    - (5 comma 6)

The graph of a system of linear equations is shown. What is the solution (x, y) to the system?

A. $(2, 3)$

B. $3, 4$

C. $(4, 5)$

D. $(5, 6)$